Mirror, Mirror on the Wall: Can VLM Agents Tell Who They Are at All?

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Abstract

In the animal kingdom, mirror self-recognition is a canonical probe of higher-order cognition, emerging only in some species. We ask whether an analogous functional capability emerges in embodied vision-language model (VLM) agents: *can they recognize themselves in a mirror?* We introduce a controlled 3D benchmark where a first-person VLM agent must infer a hidden body attribute from its reflection and select the matching target, while avoiding self–other misattribution. To separate mirror-grounded self-identification from shortcuts, we test mirror removal, misleading cues, and occluded reflections. We also evaluate the decision process through mirror seeking, temporal ordering, self-attribution, and reasoning–action consistency. Our experiments show that mirror-based self-identification emerges mainly in stronger VLMs. These models can use reflected evidence for action, whereas weaker models often inspect the mirror but fail to extract self-relevant information or misattribute their reflection. Language–vision conflict further shows that self-referential language alone is not evidence of grounded self-identification. Overall, mirror-based evaluation provides a diagnostic for whether embodied self-grounding is causally rooted in perception and action rather than priors, prompt compliance, or confabulation.

Keywords: Vision-Language Agents, Multimodal Reasoning, Self-Identification, Benchmark

1 Introduction

Vision-language models (VLMs) are increasingly used as decision-making modules for embodied agents that must perceive, reason, and act under partial observability. This makes visual grounding essential: an agent must not only describe what it sees, but use visual evidence to choose actions. Despite strong progress in recognition, instruction following, navigation, and multimodal reasoning, current VLMs still suffer from hallucination, weak visual grounding, and failures in spatial reasoning [5, 12, 26, 39]. These limitations become especially important when the object of reasoning is not an external object, but the agent itself.

Self-identification is a basic requirement for embodied agency. An agent must distinguish its own body, state, and actions from those of other agents and objects. Under egocentric perception, however, self-relevant information is often outside the camera view and cannot be accessed directly. Mirrors

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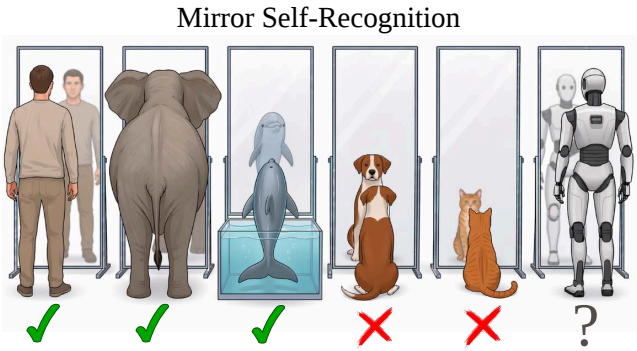


Figure 1: **Mirror self-recognition as a test of embodied self-grounding.** Across species, mirror self-recognition emerges selectively: humans, elephants, and dolphins can use mirror reflections to guide self-directed behavior, while other animals fail or rely on simpler strategies. We ask whether a similar capability emerges in vision-language model (VLM) agents. In our setting, an embodied agent must infer a hidden body attribute from a mirror and act accordingly. Success requires not only perceiving the reflection, but also attributing it to the self rather than to another entity.

provide a controlled way to test this ability: they reveal hidden body information while requiring the agent to attribute the reflected entity to itself and use that attribution for action.

Mirror self-recognition has long been studied in comparative cognition. Classic work showed that chimpanzees can use mirror reflections to guide self-directed behavior [8], and later studies reported related behavior in dolphins, elephants, and other species [21, 24]. Yet the interpretation of mirror-test success remains debated. It may reflect self-awareness, but it may also arise from sensorimotor matching, learned visual-proprioceptive associations, or task-specific strategies [3, 40]. This caution should also apply to embodied agents: fluent self-referential language or correct final actions are not sufficient evidence of grounded self-identification.

In this paper, we study *embodied self-referential grounding*: whether a VLM agent can bind first-person expressions such as "you" and "your body" to the correct physical entity and its state, and use that information to guide action (Fig. 1). Assigning the model a robot-control role does not identify which perceived body it controls or reveal its hidden attributes. Our benchmark therefore asks a top-down question: once language assigns a first-person role, can the model ground that role in perception and action? The animal mirror test inspires the experimental design, while the evaluated capability is distinct from pure self-awareness.

To study this question, we introduce a controlled benchmark[3] in which a VLM-controlled agent observes the environment from a first-person perspective and must select the target object matching its own body color. This attribute is hidden from direct view, requiring the agent to infer it from reflected visual evidence and attribute it to the body it controls. Diagnostic conditions remove the mirror, introduce misleading linguistic cues, add other agents, or present cluttered and partially occluded reflections. Together, these conditions probe whether self-reports and actions depend on relevant perceptual evidence, and expose failures of evidence use, self–other attribution, and language–vision conflict resolution.

A key aspect of our benchmark is that it evaluates both outcomes and decision processes. Target-selection accuracy indicates whether the agent acts correctly, but not whether the decision was grounded in mirror evidence. We therefore also measure mirror consultation, mirror gaze duration, temporal ordering between mirror observation and action, self-attribution in language, and confabulation. This process-level evaluation is necessary because embodied VLMs can appear competent while relying on weak grounding or non-visual shortcuts [5, 12, 26].

Our experiments show that mirror-guided self-identification can emerge in stronger VLM agents but remains fragile. Some models use reflected evidence to guide action, whereas others inspect the mirror without extracting the relevant self-attribute. We also find that self-referential language is not a reliable proxy for grounded self-identification: models often produce coherent first-person

[3]Code & Benchmark are available at https://github.com/tommyz94/self-awareness

explanations even when mirror evidence is absent, misleading, or degraded. These results suggest that mirror-based evaluation is a useful diagnostic for embodied self-grounding in VLM agents.

The benchmark is intended for evaluating embodied VLM controllers that must maintain consistent links between the body they control, their observations, their self-reports, and their actions. Its diagnostic value lies in distinguishing failures of evidence access, self–other attribution, cross-modal conflict resolution, and action selection. These distinctions are relevant when instructions refer to an agent's own state under partial observability or in environments containing several visually plausible agents.

Our contributions are as follows: (i) we frame mirror-guided self-identification as a functional embodied capability for VLM agents and introduce a controlled 3D benchmark requiring agents to infer hidden self-relevant attributes from mirror evidence and act accordingly; (ii) we design interventions that remove mirror evidence, introduce misleading language, and degrade reflections to test whether behavior depends on visual self-grounding; (iii) we propose process-level metrics covering task success, evidence seeking, temporal ordering, self-attribution, and confabulation; (iv) we show that mirror-guided self-identification emerges mainly in stronger VLMs, whereas weaker agents often consult the mirror but fail to reliably use reflected evidence, frequently mistaking themselves for another agent.

2 Related Work

Vision-Language Models for Embodied Reasoning. Vision-language models (VLMs) connect visual perception with language-conditioned reasoning and have shown strong zero-shot and few-shot performance across recognition, captioning, visual question answering, navigation, and multimodal reasoning [39]. In embodied settings, they can map visual observations and instructions into high-level plans or actions, but strong multimodal performance does not necessarily imply grounded visual reasoning: VLMs still suffer from hallucination, weak modality alignment, textual priors, and failures on spatially simple or perceptually grounded tasks [14, 16, 26, 33, 35, 41]. Recent work improves grounding through perception-language decomposition, self-rewarding, calibration, navigation history, and action-conditioned reasoning [12, 14, 33, 36, 41]. Our work shifts this focus from grounding external objects to grounding self-relevant visual information.

Multimodal Benchmarks, Shortcuts, and Process-Level Evaluation. Recent multimodal benchmarks evaluate object hallucination, referential comprehension, compositional reasoning, self-consistency, and visual-language competence [16, 30, 34, 35, 37, 41]. These benchmarks show that high accuracy can still arise from shortcuts, including language priors, dataset regularities, or memorized correlations rather than genuine image use [5]. Vision-centric benchmarks such as NaturalBench and MMStar therefore design tasks that require image-grounded reasoning [5, 13]. This issue is central to embodied self-identification: an agent may choose the correct target without using the mirror, produce self-referential language without perceptual evidence, or misattribute its reflection to another entity.

Mirror Self-Recognition in Animals and Humans. Mirror self-recognition has long been used as a behavioral probe of self-related processing in comparative cognition. The classic mark-test paradigm reported that chimpanzees can use mirrors to guide self-directed behavior [8], and later work reported related findings in chimpanzees, orangutans, dolphins, elephants, magpies, and other species, while also documenting important species-level differences [9, 17, 21, 24, 25, 29, 40]. The interpretation of these results remains contested: mirror-test success may indicate self-awareness or self-concept, but it may also reflect sensorimotor matching, kinesthetic-visual correspondence, learned associations, or task-specific strategies [2, 4, 10, 19, 20]. We adopt this cautious interpretation and test mirror-guided self-identification as a functional capacity rather than as evidence of philosophical self-awareness.

Artificial Self-Recognition and Embodied Agency. Artificial self-recognition has been studied in robotics and cognitive architectures through kinesthetic-visual matching, body-model learning, inner speech, multisensory integration, and sensorimotor contingencies [11, 23, 38]. These approaches relate to psychological and developmental accounts that distinguish first-order sensorimotor matching from more reflective self-awareness, and that emphasize action–perception coupling in distinguishing self from other [3, 18, 27, 28]. Unlike systems explicitly engineered for self-recognition, we test whether a general-purpose VLM agent can infer a hidden body attribute from first-person mirror evidence and use it for goal-directed action.

Positioning of This Work. Prior work shows that VLMs can perform well while relying on weak grounding, shortcuts, or textual priors, and that mirror self-recognition remains a contested but influential probe of self-related processing. We connect these threads by studying mirror-guided self-identification as a functional embodied capability. By measuring mirror consultation, temporal ordering, self-attribution, and confabulation, we distinguish mirror-grounded self-identification from shortcut behavior, prompt compliance, and misattribution of the agent's own reflection.

3 Setup

3.1 Motivation and Research Question

We study whether an embodied VLM agent can treat visual observations as self-referential evidence. Rather than addressing self-awareness in a philosophical sense, we focus on a functional question: can an agent infer a hidden property of its own body from a reflection and use it to guide action?

The task is designed so that self-relevant information is never directly visible from the egocentric view. The agent must instead recover it indirectly through a reflective surface, under conditions that introduce ambiguity, distractors, misleading language, or degraded visual evidence. This allows us to distinguish mirror-grounded self-identification from guessing, prompt following, scene priors, or post-hoc confabulation.

3.2 Embodied Setting and Task

The agent operates in a simulated 3D environment from a first-person perspective. At each timestep $t \in \{1, \dots, T\}$, it receives $x_t = \big(I_t, h_{<t}, p_t, q\big)$ and outputs $y_t = \big(a_t, \hat{c}_t^{\text{self}}, z_t\big)$, where I_t is the current RGB observation, $h_{<t}$ is a compact action-position history, p_t is the navigation state, q is the task instruction, and $a_t \in \{\texttt{move_forward}, \texttt{move_backward}, \texttt{turn_left}, \texttt{turn_right}, \texttt{done}\}$ is the chosen action. The variable $\hat{c}_t^{\text{self}}$ denotes the current self-attribution, while z_t denotes the task-dependent decision field: cube selection in E_1–E_4 and final self-attribution in E_5.

Across all conditions, the agent's true body color $c^\star$ is never directly visible from the egocentric view and must be inferred indirectly from perceptual evidence in the scene. The reflective mirror surface is not visually labeled, so successful behavior requires grounding self-identification in action-contingent and temporally consistent visual evidence, rather than relying only on priors or scene regularities. A visual example of all the experimental setups is shown in Figure 2, while details on the environment can be found in Appendix A.7.

We use a controlled MuJoCo environment to manipulate reflected evidence while retaining access to exact scene states and visibility annotations. Clear reflections provide a deliberately favorable perceptual condition, allowing us to study grounding failures with fewer confounds from illumination, sensor noise, or material quality. This setting supports controlled diagnosis; success does not establish robustness to photorealistic scenes or physical mirrors.

3.3 Experimental Conditions

E1: Active Mirror-Based Self-Identification. The agent is placed in a room with an unlabeled reflective surface and multiple colored cubes. Its body color is hidden from direct view. To solve the task, the agent must explore, discover the reflective evidence, infer its own body color, and navigate to the matching cube. This tests whether an embodied agent can identify itself through reflection rather than through chance or scene priors.

E2: Self-Identification Without Perceptual Evidence. This condition matches E1, but the reflective surface is removed. The agent must still select the cube corresponding to its own body color, although no visual evidence about this attribute is available. Performance should therefore collapse to chance; residual success indicates reliance on priors, prompt bias, scene regularities, or guessing.

E3: Self-Identification Under Conflicting Language. This condition matches E1, but the agent receives a misleading textual statement about its own body color that conflicts with the mirror evidence. The task tests whether the agent privileges visual self-evidence over false linguistic information, separating perceptual grounding from prompt compliance.

E4: Self–Other Disambiguation. The agent is placed in a room with one to six other agents, which may have similar or identical body colors. All agents can appear both directly and through the

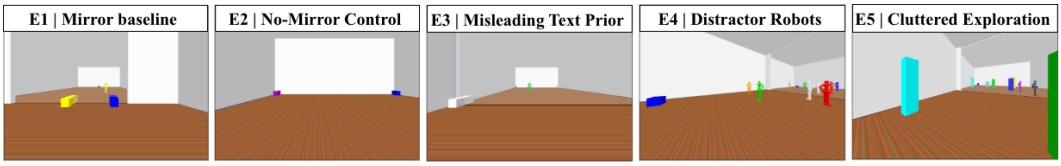


Figure 2: **Experimental settings.** Visual examples of the proposed conditions and 3D environments. At each timestep, the agent observes the current first-person view RGB frame, a compact textual history, and minimal navigation context. It then outputs a discrete navigation action, an image description, and an updated episode summary that is carried forward as context.

reflective surface. The task is to distinguish its own reflection from those of other agents and navigate to the cube matching its body color. Action–reflection synchrony provides a potential disambiguation cue, although static visual cues may also support successful decisions.

E5: Robust Open-Ended Self-Attribution Under Clutter. Unlike E_1–E_4, E5 is an open-ended exploration probe rather than a cube-selection task. The agent explores a cluttered mirrored room with distractors and must decide when to terminate with a final self-attribution. This tests whether self-identification remains possible when mirror evidence is partial, noisy, or intermittently occluded.

Formally, for each condition $e \in \{E_1, \ldots, E_5\}$, let τ denote the first timestep at which the agent emits `done`, with the convention $\tau = T$ if no `done` action is produced. Let $c^\star$ denote the true body color of the ego agent. We define the final task decision as

$$z_\tau^{(e)} = \begin{cases} \hat{c}_\tau^{\text{cube}}, & e \in \{E_1, E_2, E_3, E_4\}, \\ \hat{c}_\tau^{\text{self}}, & e = E_5, \end{cases}$$

where $\hat{c}_\tau^{\text{cube}}$ is the final selected cube color and $\hat{c}_\tau^{\text{self}}$ is the final self-color attribution.

3.4 Design Rationale

The task suite progressively probes embodied self-identification under increasing ambiguity. E1 tests the core capability, E2 removes perceptual evidence, E3 creates language–vision conflict, E4 introduces self–other ambiguity, and E5 tests robustness to degraded reflections. Together, these conditions ask whether VLM agents can recognize themselves in a mirror as a grounded functional capability, rather than merely produce correct answers or self-referential language.

3.5 Behavioral and Process-Level Measures

We evaluate both task outcomes and decision processes. Outcome metrics capture whether the agent selects the correct target, but this alone does not show whether the decision was grounded in mirror evidence. We therefore also measure whether the agent consults the mirror, whether mirror observation precedes action, whether the reflected body is attributed to the self, and whether the agent confabulates self-knowledge without evidence. Generally, for episode i, let

$$\tau_i = \min\{t : a_{i,t} = \texttt{done}\},$$

with $\tau_i = T_i$ if no `done` action is produced. Let $z_{i,\tau_i}^{(e)}$ denote the final task decision defined in Section 3.3, let $c_i^\star$ be the true ego-agent color, and let $m_{i,t} \in \{0, 1\}$ indicate whether mirror evidence is visible at timestep t.

Task Success Accuracy (TSA) measures whether the agent reaches the correct final task decision Formally:

$$\text{TSA}_i = \begin{cases} 1, & \text{if } z_{i,\tau_i}^{(e)} = c_i^\star, \\ 0, & \text{otherwise.} \end{cases}$$

Time-to-Decision (TTD) measures the number of timesteps before termination:

$$\text{TTD}_i = \tau_i.$$

It helps separating immediate biased choices from decisions preceded by exploration.

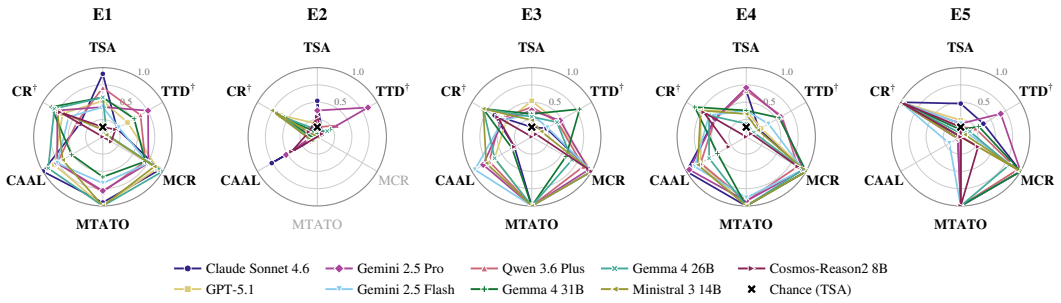


Figure 3: **Behavioral self-recognition profiles across task variants.** Model comparison across experiments using TSA, TTD, MCR, MTATO, CAAL, and CR. Higher is better on all axes; therefore, TTD and CR are reversed for visualization. The red marker on the TSA axis indicates chance level.

Mirror Consultation Rate (MCR) measures whether the agent observes mirror evidence before committing to the final decision, distinguishing mirror-based decisions from visually unsupported ones.

$$\mathrm{MCR}_i = \begin{cases} 1, & \text{if there exists } t < \tau_i \text{ such that } m_{i,t} = 1, \\ 0, & \text{otherwise.} \end{cases}$$

Mirror-Then-Action Temporal Ordering (MTATO) measures the fraction of successful episodes in which mirror consultation occurs before the final decision, testing whether correct behavior is temporally preceded by relevant evidence. For successful episodes only, we define

$$\mathrm{MTATO}_i = \begin{cases} 1, & \text{if } \mathrm{TSA}_i = 1 \text{ and there exists } t < \tau_i \text{ such that } m_{i,t} = 1, \\ 0, & \text{if } \mathrm{TSA}_i = 1 \text{ and no such } t \text{ exists.} \end{cases}$$

We then report the mean of MTATO_i over successful episodes only.

Correct Attribution At Least Once (CAAL) measures whether the agent ever states the correct self-color before termination:

$$\mathrm{CAAL}_i = \begin{cases} 1, & \text{if there exists } t < \tau_i \text{ such that } \hat{c}_{i,t}^{\mathrm{self}} = c_i^{\star}, \\ 0, & \text{otherwise.} \end{cases}$$

Confabulation Rate (CR) measures whether the agent commits to a self-attribution before observing mirror evidence. Let t_i^{claim} denote the first timestep before τ_i at which the agent outputs a self-color different from `unknown`, and let t_i^{mirror} denote the first timestep before τ_i at which mirror evidence becomes visible, whenever these timesteps exist. We define

$$\mathrm{CR}_i = \begin{cases} 1, & \text{if } t_i^{\mathrm{claim}} \text{ exists and either } t_i^{\mathrm{mirror}} \text{ does not exist} \\ & \text{or } t_i^{\mathrm{claim}} < t_i^{\mathrm{mirror}}, \\ 0, & \text{otherwise.} \end{cases}$$

Together, these binary indicators report event frequencies across applicable episodes such as behavior, guessing, prompt compliance, and post-hoc explanation. Extreme values can reflect consistent behavior or experimental design, such as MCR = 0 in E2. Their joint interpretation remains informative: a model may reach a MCR of 1.0 but have a low or mid TSA value in E1, separating evidence exposure from successful task completion.

4 Results

We analyze the results with respect to the diagnostic questions and hypotheses defined in Sec. 3.3, as shown in Table 1. The central goal is not only to determine whether an agent reaches the correct final task decision but also to ensure that correct behavior is supported by mirror-grounded self-identification. We therefore interpret the results through the three metric groups introduced in Sec. 3.5: task-success metrics, mirror-interaction quality metrics, and text-based metrics. This distinction is crucial: an agent may succeed by chance or shortcut, may inspect the mirror without extracting the relevant self-attribute, or may produce fluent self-referential language without perceptual grounding. For additional technical details, extended metrics, and qualitative examples, refer to Appendix A.

We selected nine different VLMs to span the current frontier of multimodal agents across major providers (Anthropic [1], Google [6], OpenAI [7], Mistral [15], Qwen [31], Cosmos-Reason2 [22])

and a wide range of capability tiers, from compact open-weight models to flagship proprietary systems. This spread allows us to test whether mirror-guided self-identification scales with model strength or emerges only beyond a capability threshold. All VLM inference was performed via provider APIs; the only local compute requirement was a single workstation (128 GB RAM, NVIDIA RTX 2040 8 GB) used to run the MuJoCo [32] simulation and orchestrate API calls.

E1: Can an Embodied Agent Discover Who It Is by Looking at Its Reflection? E1 evaluates the core capability targeted by the benchmark: whether an embodied VLM agent can discover a reflective surface, infer its hidden body color from the reflected body, attribute that reflection to itself, and use the inferred self-attribute to guide navigation. The results in Table 1 and Figure 3 provide evidence that this capability can emerge, but only for some models and not as a uniform property of VLM agents.

From the perspective of task success, `Claude Sonnet 4.6` provides the strongest evidence of mirror-based self-identification, achieving TSA $= 0.905 \pm 0.066$. `Qwen 3.6 Plus` also performs strongly, with TSA $= 0.714 \pm 0.101$. These results show that some agents can solve the basic mirror-guided self-identification task. However, the capability is highly model-dependent: `Ministral 3 14B` reaches only TSA $= 0.095 \pm 0.066$, indicating that the task remains difficult even when the mirror is available.

The mirror-interaction metrics reveal the main diagnostic dissociation. High mirror consultation does not guarantee grounded self-identification. `Gemma4 26B` consults the mirror in every episode (MCR $= 1.000 \pm 0.000$) and has perfect temporal ordering (MTATO $= 1.000 \pm 0.000$), but reaches only TSA $= 0.571 \pm 0.111$. The same pattern is more pronounced for `Ministral 3 14B`, which has high MCR and perfect MTATO but very low TSA. Thus, the relevant failure is not merely failing to find the mirror, but failing to extract and use the reflected self-attribute.

Text-based metrics further caution against treating self-referential language as evidence of self-grounding. `Claude Sonnet 4.6` achieves perfect CAAL but also a relatively high CR of 0.619 ± 0.109, while `Qwen 3.6 Plus` combines strong TSA with CR $= 0.571 \pm 0.111$. Overall, E1 partially supports the hypothesis: some agents can use reflected evidence to infer who they are, but neither mirror consultation nor self-attribution alone is sufficient to establish grounded self-identification.

E2: What Does the Agent Do When the Mirror Is Gone? E2 serves as a negative control by removing the mirror while leaving the task otherwise unchanged. Under the hypothesis in Sec. 3.3, performance should collapse when self-referential visual evidence is unavailable; any residual success should be interpreted as shortcut behavior, prior bias, or guessing rather than grounded self-identification.

The task-success metrics in Table 1 and Figure 3 broadly support this interpretation. Most models show low TSA when the mirror is removed: `Ministral 3 14B` reaches TSA $= 0.000 \pm 0.000$, `Gemma4 31B` reaches 0.048 ± 0.048, `Gemma4 26B` reaches 0.095 ± 0.066, and `Qwen 3.6 Plus` reaches 0.143 ± 0.078. This drop relative to E1 supports the causal role of mirror evidence. However, `Claude Sonnet 4.6` retains TSA $= 0.524 \pm 0.112$, and `Gemini 2.5 Pro` retains TSA $= 0.381 \pm 0.109$.

Mirror interaction collapses completely, with MCR $= 0$ for all models and MTATO correspondingly zero or undefined, confirming that the relevant perceptual channel has been removed. More interestingly, TTD reveals two distinct response patterns: some models commit quickly despite lacking evidence, most clearly `Gemini 2.5 Pro` (TTD $= 15.333 \pm 2.493$, CR $= 0.952 \pm 0.048$), whereas others continue exploring much longer under uncertainty, such as `GPT 5.1` (TTD $= 98.476 \pm 0.742$). Several agents continue to assert self-color identifications even when they lack perceptual access to the relevant visual evidence. `Claude Sonnet 4.6` has CAAL $= 0.762 \pm 0.095$ but CR. $= 1.000 \pm 0.000$, while `Gemini 2.5 Pro` and `Gemini 2.5 Flash` reach CR. $= 0.952 \pm 0.048$ and 0.905 ± 0.066, respectively. Overall, E2 exposes unsupported self-attributions in the absence of mirror evidence: fluent self-referential language and occasional correct decisions do not establish grounded self-identification.

E3: Can the Agent Trust Its Eyes Over a False Belief About Itself? E3 introduces a misleading textual statement about the agent's body color while preserving access to the mirror. This condition tests whether the agent privileges reflected visual evidence over an incorrect linguistic prior. The results in Table 1 show that this ability is limited and highly model-dependent.

Task success is substantially lower than in the clean mirror condition for several models. The best-performing model is `GPT 5.1`, with TSA $= 0.524 \pm 0.112$, followed by `Qwen 3.6 Plus` with TSA $= 0.429 \pm 0.111$ and `Gemini 2.5 Pro` with TSA $= 0.381 \pm 0.109$. The most striking degradation is observed for `Claude Sonnet 4.6`, which drops from TSA $= 0.905 \pm 0.066$ in E1 to 0.143 ± 0.078 in E3. Thus, success in the clean mirror setting does not imply robustness to language–vision conflict.

The mirror-interaction metrics show that this degradation is not due to failure to consult the mirror. Several models have MCR $= 1.000 \pm 0.000$, including `Claude Sonnet 4.6`, `Gemini 2.5 Flash`, `GPT 5.1`, and `Ministral 3 14B`. MTATO is also perfect for all models with defined values. The main failure mode is therefore epistemic rather than exploratory: agents often access the relevant visual evidence but do not reliably resolve the conflict in favor of perception.

The text-based metrics reveal a further action–language dissociation. `Gemini 2.5 Flash` reaches CAAL $= 0.952 \pm 0.048$ but only TSA $= 0.333 \pm 0.105$, and `Ministral 3 14B` reaches CAAL $= 0.762 \pm 0.095$ but TSA $= 0.238 \pm 0.095$. These results show that correctly stating the self-color does not guarantee a correct embodied action. E3, therefore, only partially supports the hypothesis: agents can sometimes trust visual evidence over a false textual belief, but this ability is unstable and often breaks between verbal attribution and navigation.

E4: Can the Agent Tell Which Reflection Is Itself When Others Look Similar? E4 probes self–other disambiguation. The hypothesis is that successful agents should go beyond static appearance matching and use action-contingent evidence to identify which reflected body corresponds to themselves. The results in Table 1 and Figure 3 suggest that some models can handle this ambiguity, but the capability remains fragile.

In terms of task success, `Gemini 2.5 Pro` performs best, with TSA $= 0.714 \pm 0.101$. `Claude Sonnet 4.6` and `Qwen 3.6 Plus` both reach TSA $= 0.667 \pm 0.105$. These results indicate that self–other ambiguity does not fully prevent mirror-guided action for stronger models. However, the performance gap across models remains large: `Gemma4 26B` reaches only TSA $= 0.190 \pm 0.088$ despite high mirror consultation.

The mirror-interaction metrics again separate evidence seeking from self-identification. MCR is high for all models, and MTATO is generally high, but TSA varies widely. `Gemma4 26B` has MCR $= 0.952 \pm 0.048$ and MTATO $= 1.000 \pm 0.000$, but low TSA. `Ministral 3 14B` has MCR $= 1.000 \pm 0.000$ and MTATO $= 1.000 \pm 0.000$, but TSA $= 0.333 \pm 0.105$. These failures suggest that agents often inspect the mirror before acting, yet fail to determine which reflected entity is causally linked to themselves.

Text-based metrics remain only partially aligned with behavior. Stronger models often achieve high CAAL, including `Claude Sonnet 4.6` with CAAL $= 1.000 \pm 0.000$ and `Gemini 2.5 Pro` with CAAL $= 0.952 \pm 0.048$. Overall, E4 partially supports the hypothesis: stronger agents can sometimes identify which reflection is themselves, but self–other disambiguation remains brittle and cannot be inferred from mirror consultation alone. A trajectory audit found qualifying pre/post-action ego-reflection exposure in only 8/168 episodes and did not establish synchrony-based attribution. E4 success therefore does not demonstrate the use of action–reflection synchrony; static cues remain plausible.

E5: Can the Agent Still Recognize Itself Through a Partially Occluded Reflection?
E5 tests robustness under degraded mirror evidence. If agents relied on robust causal or temporal cues rather than static visual matching alone, performance should degrade gradually under occlusion. The results in Table 1 instead show a broad collapse, indicating that current mirror-guided self-identification remains brittle.

Task success is substantially reduced across models. `Claude Sonnet 4.6` remains the strongest model but drops to TSA $= 0.476 \pm 0.112$. `GPT 5.1` follows with TSA $= 0.238 \pm 0.095$, while most other models are near the bottom of the range: `Gemma4 26B` reaches TSA $= 0.048 \pm 0.048$, `Gemma4 31B` reaches 0.095 ± 0.066, `Qwen 3.6 Plus` reaches 0.143 ± 0.078, and `Ministral 3 14B` reaches 0.000 ± 0.000.

The mirror-interaction metrics show that low task success often coexists with frequent mirror exposure. MCR remains high for most models despite their low TSA. These results show that frequent mirror exposure alone does not ensure successful self-identification.

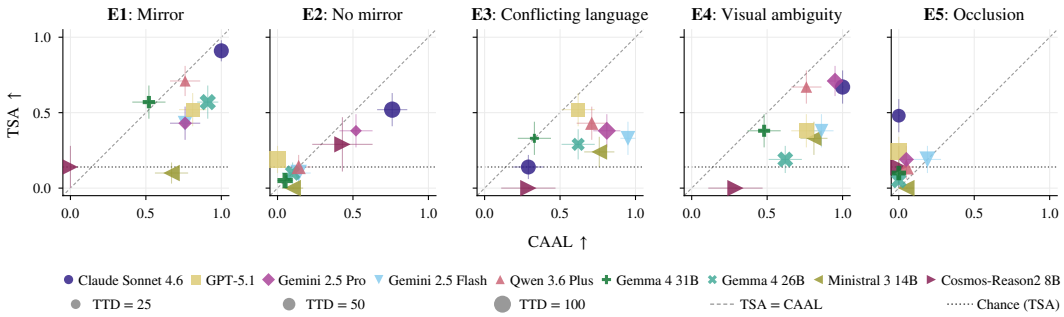


Figure 4: **Task success versus self-attribution across experiments.** Each point corresponds to one model in one experiment, with the horizontal axis showing CAAL and the vertical axis showing TSA. The dashed diagonal marks $\mathrm{TSA} = \mathrm{CAAL}$. Marker size encodes relative TTD, with larger markers indicating slower decisions. Points below the diagonal indicate cases in which correct self-attribution is not consistently translated into correct embodied action.

The text-based metrics show a distinctive form of failure. CAAL collapses to zero or near zero for most models, including `Claude Sonnet 4.6`, `Gemma4 26B`, `Gemma4 31B`, and `GPT 5.1`, while CR is near zero across almost all models. This should not be interpreted as improved grounding; rather, many agents fail to make a concrete self-attribution at all under extended uncertainty. E5 therefore demonstrates that mirror access alone is insufficient for robust self-identification in open-ended settings. Although agents continue to seek mirror evidence, they generally fail to use it effectively when the reflection is partially occluded, thereby rejecting the robustness hypothesis. Finally, Figure 4 highlights the gap between correct self-attribution and final task success; complete results and additional analyzes are provided in the supplementary material.

5 Conclusions

We introduced a controlled benchmark for functional mirror-guided self-identification in embodied VLM agents. Our results show that this capability can emerge, but it remains partial, fragile, and strongly model-dependent. Performance is highest in the clean mirror condition, but drops when mirror evidence is removed, when language conflicts with perception, and under self–other ambiguity or cluttered reflected scenes. This supports the causal role of reflected visual evidence while also showing that success in the basic condition should not be overinterpreted as robust self-awareness-like behavior.

Across conditions, the main pattern is a dissociation between accessing evidence and using it. Many agents consult the mirror, often before acting, yet still fail to extract the relevant self-attribute or translate it into the correct final decision. Language introduces a second dissociation: agents can produce plausible or even correct self-attributions without reliable perceptual grounding, as shown by confabulation in the no-mirror control and by the gap between CAAL and TSA under misleading language. Overall, the strongest evidence for mirror-grounded self-identification appears only when task success, mirror consultation, temporal ordering, and correct self-attribution align.

Limitations and Future Work. This benchmark evaluates embodied self-referential grounding through mirror-guided tasks; it does not assess philosophical self-awareness. VLMs may partly rely on pretraining priors about mirrors and reflections. We use an unlabeled reflective surface without a canonical mirror frame to limit explicit semantic cues, but this does not eliminate learned visual priors or alternative strategies. The evaluation is restricted to body color, discrete navigation actions, and first-person RGB observations with textual history, without continuous sensorimotor coupling. Generalization to other attributes remains untested, and the current evidence does not establish that successful agents exploit action–reflection synchrony. Each primary model–condition pair contains 21 episodes, limiting estimate precision and the strength of comparisons between models with similar scores. Sensitivity to the fixed decoding configuration was not evaluated. Furthermore, E5 changes both scene complexity and the task prompt, so differences from E1 cannot be attributed to occlusion alone. Success in this controlled simulation also does not establish robustness to physical mirrors or real-world sensory conditions.

Future work should vary self-attributes, directly manipulate action–reflection synchrony, and isolate visual degradation under matched task instructions. Video-based observations, collaborative multi-

Model	TSA ↑	TTD ↓	MCR ↑	MTATO ↑	CAAL ↑	CR ↓
E1: Active mirror-based self-identification						
Cosmos-Reason2-8B	0.14 ± 0.14	78.86 ± 14.47	0.14 ± 0.14	0.00 ± 0.00	0.00 ± 0.00	0.29 ± 0.18
Ministral 3 14B	0.10 ± 0.07	94.43 ± 3.56	0.86 ± 0.08	$\mathbf{1.00 \pm 0.00}$	0.67 ± 0.11	0.24 ± 0.10
Gemma4 26B	0.57 ± 0.11	93.43 ± 4.08	$\mathbf{1.00 \pm 0.00}$	$\mathbf{1.00 \pm 0.00}$	$\underline{0.91 \pm 0.07}$	$\underline{0.19 \pm 0.09}$
Gemma4 31B	0.57 ± 0.11	47.67 ± 8.15	0.71 ± 0.10	0.58 ± 0.15	0.52 ± 0.11	$\mathbf{0.14 \pm 0.08}$
Gemini 2.5 Flash	0.43 ± 0.11	73.43 ± 7.63	0.76 ± 0.10	0.67 ± 0.17	0.76 ± 0.10	0.43 ± 0.11
Gemini 2.5 Pro	0.43 ± 0.11	$\mathbf{24.48 \pm 4.27}$	0.76 ± 0.10	0.78 ± 0.15	0.76 ± 0.10	0.24 ± 0.10
Qwen 3.6 Plus	$\underline{0.71 \pm 0.10}$	$\underline{37.10 \pm 6.33}$	0.71 ± 0.10	0.80 ± 0.11	0.76 ± 0.10	0.57 ± 0.11
GPT 5.1	0.52 ± 0.11	58.71 ± 8.24	0.91 ± 0.07	$\mathbf{1.00 \pm 0.00}$	0.81 ± 0.09	0.33 ± 0.11
Claude Sonnet 4.6	$\mathbf{0.91 \pm 0.07}$	76.33 ± 6.24	$\underline{0.95 \pm 0.05}$	$\underline{0.95 \pm 0.05}$	$\mathbf{1.00 \pm 0.00}$	0.62 ± 0.11
E2: Self-identification without mirror evidence						
Cosmos-Reason2-8B	0.29 ± 0.18	91.86 ± 8.14	$\mathbf{0.00 \pm 0.00}$	$\mathbf{0.00 \pm 0.00}$	0.43 ± 0.20	0.86 ± 0.14
Ministral 3 14B	0.00 ± 0.00	95.62 ± 3.06	$\mathbf{0.00 \pm 0.00}$	–	0.10 ± 0.07	$\mathbf{0.24 \pm 0.10}$
Gemma4 26B	0.10 ± 0.07	83.43 ± 7.49	$\mathbf{0.00 \pm 0.00}$	$\mathbf{0.00 \pm 0.00}$	0.10 ± 0.07	0.43 ± 0.11
Gemma4 31B	0.05 ± 0.05	78.33 ± 7.10	$\mathbf{0.00 \pm 0.00}$	$\mathbf{0.00 \pm 0.00}$	0.05 ± 0.05	$\underline{0.33 \pm 0.11}$
Gemini 2.5 Flash	0.10 ± 0.07	94.67 ± 3.94	$\mathbf{0.00 \pm 0.00}$	$\mathbf{0.00 \pm 0.00}$	0.14 ± 0.08	0.91 ± 0.07
Gemini 2.5 Pro	$\underline{0.38 \pm 0.11}$	$\mathbf{15.33 \pm 2.49}$	$\mathbf{0.00 \pm 0.00}$	$\mathbf{0.00 \pm 0.00}$	$\underline{0.52 \pm 0.11}$	0.95 ± 0.05
Qwen 3.6 Plus	0.14 ± 0.08	$\underline{67.67 \pm 8.94}$	$\mathbf{0.00 \pm 0.00}$	$\mathbf{0.00 \pm 0.00}$	0.14 ± 0.08	0.76 ± 0.10
GPT 5.1	0.19 ± 0.09	98.48 ± 0.74	$\mathbf{0.00 \pm 0.00}$	$\mathbf{0.00 \pm 0.00}$	0.00 ± 0.00	0.38 ± 0.11
Claude Sonnet 4.6	$\mathbf{0.52 \pm 0.11}$	91.67 ± 2.55	$\mathbf{0.00 \pm 0.00}$	$\mathbf{0.00 \pm 0.00}$	$\mathbf{0.76 \pm 0.10}$	1.00 ± 0.00
E3: Self-identification under conflicting linguistic information						
Cosmos-Reason2-8B	0.00 ± 0.00	92.57 ± 7.43	$\mathbf{1.00 \pm 0.00}$	–	0.29 ± 0.18	0.43 ± 0.20
Ministral 3 14B	0.24 ± 0.10	84.33 ± 5.83	$\mathbf{1.00 \pm 0.00}$	$\mathbf{1.00 \pm 0.00}$	0.76 ± 0.10	$\mathbf{0.19 \pm 0.09}$
Gemma4 26B	0.29 ± 0.10	57.90 ± 10.07	0.67 ± 0.11	$\mathbf{1.00 \pm 0.00}$	0.62 ± 0.11	$\mathbf{0.19 \pm 0.09}$
Gemma4 31B	0.33 ± 0.11	$\mathbf{20.00 \pm 7.44}$	0.57 ± 0.11	$\mathbf{1.00 \pm 0.00}$	0.33 ± 0.11	$\mathbf{0.19 \pm 0.09}$
Gemini 2.5 Flash	0.33 ± 0.11	73.71 ± 7.21	$\mathbf{1.00 \pm 0.00}$	$\mathbf{1.00 \pm 0.00}$	$\mathbf{0.95 \pm 0.05}$	0.52 ± 0.11
Gemini 2.5 Pro	0.38 ± 0.11	$\underline{52.81 \pm 9.18}$	$\underline{0.95 \pm 0.05}$	$\mathbf{1.00 \pm 0.00}$	$\underline{0.81 \pm 0.09}$	0.52 ± 0.11
Qwen 3.6 Plus	$\underline{0.43 \pm 0.11}$	57.10 ± 8.98	0.86 ± 0.08	$\mathbf{1.00 \pm 0.00}$	0.71 ± 0.10	0.48 ± 0.11
GPT 5.1	$\mathbf{0.52 \pm 0.11}$	57.19 ± 7.84	$\mathbf{1.00 \pm 0.00}$	$\mathbf{1.00 \pm 0.00}$	0.62 ± 0.11	0.57 ± 0.11
Claude Sonnet 4.6	0.14 ± 0.08	75.67 ± 7.57	$\mathbf{1.00 \pm 0.00}$	$\mathbf{1.00 \pm 0.00}$	0.29 ± 0.10	$\underline{0.38 \pm 0.11}$
E4: Self–other disambiguation under visual ambiguity						
Cosmos-Reason2-8B	0.00 ± 0.00	91.43 ± 5.57	0.86 ± 0.14	–	0.29 ± 0.18	0.29 ± 0.18
Ministral 3 14B	0.33 ± 0.11	80.57 ± 7.17	$\mathbf{1.00 \pm 0.00}$	$\mathbf{1.00 \pm 0.00}$	0.81 ± 0.09	$\underline{0.24 \pm 0.10}$
Gemma4 26B	0.19 ± 0.09	87.52 ± 4.15	$\underline{0.95 \pm 0.05}$	$\mathbf{1.00 \pm 0.00}$	0.62 ± 0.11	0.38 ± 0.11
Gemma4 31B	0.38 ± 0.11	45.05 ± 7.32	$\mathbf{1.00 \pm 0.00}$	$\mathbf{1.00 \pm 0.00}$	0.48 ± 0.11	$\mathbf{0.14 \pm 0.08}$
Gemini 2.5 Flash	0.38 ± 0.11	62.76 ± 8.46	$\underline{0.95 \pm 0.05}$	0.88 ± 0.13	0.86 ± 0.08	0.38 ± 0.11
Gemini 2.5 Pro	$\mathbf{0.71 \pm 0.10}$	$\underline{42.57 \pm 6.03}$	$\underline{0.95 \pm 0.05}$	$\underline{0.93 \pm 0.07}$	$\underline{0.95 \pm 0.05}$	0.48 ± 0.11
Qwen 3.6 Plus	$\underline{0.67 \pm 0.11}$	$\mathbf{40.43 \pm 6.39}$	0.86 ± 0.08	$\underline{0.93 \pm 0.07}$	0.76 ± 0.10	0.48 ± 0.11
GPT 5.1	0.38 ± 0.11	75.29 ± 6.11	$\mathbf{1.00 \pm 0.00}$	$\mathbf{1.00 \pm 0.00}$	0.76 ± 0.10	$\underline{0.24 \pm 0.10}$
Claude Sonnet 4.6	$\underline{0.67 \pm 0.11}$	78.67 ± 6.75	$\mathbf{1.00 \pm 0.00}$	$\mathbf{1.00 \pm 0.00}$	$\mathbf{1.00 \pm 0.00}$	0.48 ± 0.11
E5: Robust self-identification under occlusion						
Cosmos-Reason2-8B	0.14 ± 0.14	100.00 ± 0.00	0.29 ± 0.18	$\mathbf{1.00 \pm 0.00}$	0.00 ± 0.00	$\mathbf{0.00 \pm 0.00}$
Ministral 3 14B	0.00 ± 0.00	93.43 ± 4.14	$\mathbf{1.00 \pm 0.00}$	–	$\underline{0.05 \pm 0.05}$	$\mathbf{0.00 \pm 0.00}$
Gemma4 26B	0.05 ± 0.05	84.57 ± 6.86	0.81 ± 0.09	$\mathbf{1.00 \pm 0.00}$	0.00 ± 0.00	$\underline{0.05 \pm 0.05}$
Gemma4 31B	0.10 ± 0.07	70.00 ± 6.93	$\mathbf{1.00 \pm 0.00}$	$\mathbf{1.00 \pm 0.00}$	0.00 ± 0.00	$\mathbf{0.00 \pm 0.00}$
Gemini 2.5 Flash	0.19 ± 0.09	88.86 ± 3.90	$\mathbf{1.00 \pm 0.00}$	$\mathbf{1.00 \pm 0.00}$	$\mathbf{0.19 \pm 0.09}$	$\mathbf{0.00 \pm 0.00}$
Gemini 2.5 Pro	0.19 ± 0.09	$\mathbf{33.19 \pm 5.21}$	$\mathbf{1.00 \pm 0.00}$	$\mathbf{1.00 \pm 0.00}$	$\underline{0.05 \pm 0.05}$	$\mathbf{0.00 \pm 0.00}$
Qwen 3.6 Plus	0.14 ± 0.08	75.57 ± 8.12	$\underline{0.95 \pm 0.05}$	$\mathbf{1.00 \pm 0.00}$	$\underline{0.05 \pm 0.05}$	$\mathbf{0.00 \pm 0.00}$
GPT 5.1	$\underline{0.24 \pm 0.10}$	99.19 ± 0.15	$\underline{0.95 \pm 0.05}$	$\mathbf{1.00 \pm 0.00}$	0.00 ± 0.00	$\mathbf{0.00 \pm 0.00}$
Claude Sonnet 4.6	$\mathbf{0.48 \pm 0.11}$	$\underline{62.10 \pm 3.43}$	$\mathbf{1.00 \pm 0.00}$	$\mathbf{1.00 \pm 0.00}$	0.00 ± 0.00	$\mathbf{0.00 \pm 0.00}$

Table 1: **Experimental results across all defined settings**. Values are mean ± standard error. Best results are in bold, second-best results are underlined.

agent tasks, action-conditioned world models, and evaluations in photorealistic simulations or physical robots could further test the robustness and generality of embodied self-referential grounding.

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A Additional Results

A.1 Auxiliary Metrics and Relative Results

In addition to the core metrics reported in the main paper, we retain four auxiliary metrics that provide additional diagnostic information about completion policy, persistence of mirror exposure, and belief revision. These metrics were excluded from the main text because they are either partially conditioned or secondary to the core task-level measures, but they remain useful in the supplementary analyses below.

Using the notation introduced in Sec. A.6, let T_i denote the length of episode i, τ_i the first timestep at which the agent outputs `done` when such a timestep exists, $c_i^\star$ the ground-truth body color, $\hat{c}_{i,t}$ the self-color reported at timestep t, and $m_{i,t} \in \{0, 1\}$ whether mirror evidence is visible at timestep t.

Completion-Conditioned Task Success. Let $C_i = \mathbf{1}[\tau_i \text{ exists}]$ denote the indicator that episode i terminates with `done`. The completion-conditioned success rate is

$$\mathrm{TSA}_{\mathrm{complete}} = \frac{\sum_i C_i \, \mathrm{TSA}_i}{\sum_i C_i},$$

and is reported in tables as `TSA-C`. This quantity separates incorrect final decisions from failure to terminate appropriately.

Mirror Gaze Duration. Mirror Gaze Duration is defined at the episode level as

$$\mathrm{MGD}_i = \sum_{t=1}^{T_i} m_{i,t},$$

that is, the number of timesteps in which mirror evidence is visible during the trajectory. It measures persistence of mirror exposure rather than successful use of that exposure.

Self-Correction. Let

$$t_i^{\mathrm{mirror}} = \min\{t : m_{i,t} = 1\}$$

denote the first timestep at which mirror evidence becomes available, whenever such a timestep exists. Let

$$t_i^{\mathrm{guess}} = \min\{t < t_i^{\mathrm{mirror}} : \hat{c}_{i,t} \notin \{\texttt{unknown}, \texttt{null}, \varnothing\}\}$$

denote the first committed pre-mirror self-attribution, when it exists. The self-correction applicability indicator is

$$A_i = \mathbf{1}\left[t_i^{\mathrm{mirror}} \text{ exists},\ t_i^{\mathrm{guess}} \text{ exists},\ \hat{c}_{i,t_i^{\mathrm{guess}}} \neq c_i^\star\right].$$

For applicable episodes, self-correction is defined as

$$\mathrm{SC}_i = \mathbf{1}\left[\exists t \in [t_i^{\mathrm{mirror}}, \tilde{\tau}_i] \text{ such that } \hat{c}_{i,t} = c_i^\star\right],$$

where $\tilde{\tau}_i = \tau_i$ if the episode terminates and $\tilde{\tau}_i = T_i$ otherwise.

Applicability Rate. Because self-correction is meaningful only for episodes in which a wrong pre-mirror self-attribution is made, we also report the self-correction applicability rate

$$\mathrm{AR}_{\mathrm{SC}} = \frac{1}{N} \sum_i A_i,$$

together with the correction rate

$$\mathrm{SC} = \frac{\sum_i A_i \, \mathrm{SC}_i}{\sum_i A_i}.$$

Thus, $\mathrm{AR}_{\mathrm{SC}}$ measures how often the model enters a state where correction is possible, whereas SC measures how often it successfully revises the initial wrong self-attribution once mirror evidence becomes available.

Table 2 reports an extended view of the results, combining selected core metrics with the auxiliary metrics introduced above. The table is organized to highlight completion policy, persistence of mirror exposure, revision behavior, and language-side diagnostics.

Model	TSA ↑	TSA-C ↑	MCR ↑	MGD ↑	CAAL ↑	SC ↑	AR_{SC} ↑
E1: Active mirror-based self-identification							
Cosmos-Reason2-8B	0.14 ± 0.14	0.33 ± 0.33	0.14 ± 0.14	6.29 ± 6.29	0.00 ± 0.00	–	0.00 ± 0.00
Ministral 3 14B	0.10 ± 0.07	0.67 ± 0.33	0.86 ± 0.08	11.81 ± 2.79	0.67 ± 0.11	$\mathbf{1.00 \pm 0.00}$	0.10 ± 0.06
Gemma4 26B	0.57 ± 0.11	0.80 ± 0.11	$\mathbf{1.00 \pm 0.00}$	$\mathbf{28.48 \pm 4.53}$	0.90 ± 0.07	$\mathbf{1.00 \pm 0.00}$	0.14 ± 0.08
Gemma4 31B	0.57 ± 0.11	0.80 ± 0.11	0.71 ± 0.10	7.48 ± 2.14	0.52 ± 0.11	–	0.00 ± 0.00
Gemini 2.5 Flash	0.43 ± 0.11	0.60 ± 0.13	0.76 ± 0.10	10.76 ± 2.41	0.76 ± 0.10	$\mathbf{1.00 \pm 0.00}$	0.05 ± 0.05
Gemini 2.5 Pro	0.43 ± 0.11	0.43 ± 0.11	0.76 ± 0.10	6.86 ± 1.59	0.76 ± 0.10	–	0.00 ± 0.00
Qwen 3.6 Plus	0.71 ± 0.10	0.79 ± 0.10	0.71 ± 0.10	6.57 ± 1.62	0.76 ± 0.10	0.75 ± 0.25	$\mathbf{0.19 \pm 0.09}$
GPT 5.1	0.52 ± 0.11	0.52 ± 0.11	0.90 ± 0.07	12.00 ± 2.30	0.81 ± 0.09	0.00 ± 0.00	0.05 ± 0.05
Claude Sonnet 4.6	$\mathbf{0.90 \pm 0.07}$	$\mathbf{0.90 \pm 0.07}$	0.95 ± 0.05	21.67 ± 4.55	$\mathbf{1.00 \pm 0.00}$	$\mathbf{1.00 \pm 0.00}$	0.14 ± 0.08
E2: Self-identification without mirror evidence							
Cosmos-Reason2-8B	0.29 ± 0.18	$\mathbf{1.00 \pm 0.00}$	$\mathbf{0.00 \pm 0.00}$	$\mathbf{0.00 \pm 0.00}$	0.43 ± 0.20	–	$\mathbf{0.00 \pm 0.00}$
Ministral 3 14B	0.00 ± 0.00	0.00 ± 0.00	$\mathbf{0.00 \pm 0.00}$	$\mathbf{0.00 \pm 0.00}$	0.10 ± 0.07	–	$\mathbf{0.00 \pm 0.00}$
Gemma4 26B	0.10 ± 0.07	0.29 ± 0.18	$\mathbf{0.00 \pm 0.00}$	$\mathbf{0.00 \pm 0.00}$	0.10 ± 0.07	–	$\mathbf{0.00 \pm 0.00}$
Gemma4 31B	0.05 ± 0.05	0.25 ± 0.25	$\mathbf{0.00 \pm 0.00}$	$\mathbf{0.00 \pm 0.00}$	0.05 ± 0.05	–	$\mathbf{0.00 \pm 0.00}$
Gemini 2.5 Flash	0.10 ± 0.07	0.15 ± 0.10	$\mathbf{0.00 \pm 0.00}$	$\mathbf{0.00 \pm 0.00}$	0.14 ± 0.08	–	$\mathbf{0.00 \pm 0.00}$
Gemini 2.5 Pro	0.38 ± 0.11	0.40 ± 0.11	$\mathbf{0.00 \pm 0.00}$	$\mathbf{0.00 \pm 0.00}$	0.52 ± 0.11	–	$\mathbf{0.00 \pm 0.00}$
Qwen 3.6 Plus	0.14 ± 0.08	0.33 ± 0.17	$\mathbf{0.00 \pm 0.00}$	$\mathbf{0.00 \pm 0.00}$	0.14 ± 0.08	–	$\mathbf{0.00 \pm 0.00}$
GPT 5.1	0.19 ± 0.09	0.19 ± 0.09	$\mathbf{0.00 \pm 0.00}$	$\mathbf{0.00 \pm 0.00}$	0.00 ± 0.00	–	$\mathbf{0.00 \pm 0.00}$
Claude Sonnet 4.6	$\mathbf{0.52 \pm 0.11}$	$\mathbf{0.52 \pm 0.11}$	$\mathbf{0.00 \pm 0.00}$	$\mathbf{0.00 \pm 0.00}$	$\mathbf{0.76 \pm 0.10}$	–	$\mathbf{0.00 \pm 0.00}$
E3: Self-identification under conflicting linguistic information							
Cosmos-Reason2-8B	0.00 ± 0.00	0.00 ± 0.00	$\mathbf{1.00 \pm 0.00}$	26.57 ± 7.31	0.29 ± 0.18	0.33 ± 0.33	0.43 ± 0.20
Ministral 3 14B	0.24 ± 0.10	$\mathbf{0.83 \pm 0.17}$	$\mathbf{1.00 \pm 0.00}$	$\mathbf{38.81 \pm 3.06}$	0.76 ± 0.10	0.67 ± 0.33	0.14 ± 0.08
Gemma4 26B	0.29 ± 0.10	0.60 ± 0.16	0.67 ± 0.11	23.05 ± 4.49	0.62 ± 0.11	$\mathbf{1.00 \pm 0.00}$	0.19 ± 0.09
Gemma4 31B	0.33 ± 0.11	0.78 ± 0.15	0.57 ± 0.11	9.67 ± 3.26	0.33 ± 0.11	0.67 ± 0.33	0.14 ± 0.08
Gemini 2.5 Flash	0.33 ± 0.11	0.47 ± 0.13	$\mathbf{1.00 \pm 0.00}$	38.24 ± 4.00	$\mathbf{0.95 \pm 0.05}$	$\mathbf{1.00 \pm 0.00}$	0.52 ± 0.11
Gemini 2.5 Pro	0.38 ± 0.11	0.62 ± 0.14	0.95 ± 0.05	33.33 ± 5.95	0.81 ± 0.09	0.80 ± 0.13	0.48 ± 0.11
Qwen 3.6 Plus	0.43 ± 0.11	0.82 ± 0.12	0.86 ± 0.08	32.76 ± 5.55	0.71 ± 0.10	0.80 ± 0.13	0.48 ± 0.11
GPT 5.1	$\mathbf{0.52 \pm 0.11}$	0.52 ± 0.11	$\mathbf{1.00 \pm 0.00}$	27.81 ± 4.02	0.62 ± 0.11	0.50 ± 0.15	$\mathbf{0.57 \pm 0.11}$
Claude Sonnet 4.6	0.14 ± 0.08	0.15 ± 0.08	$\mathbf{1.00 \pm 0.00}$	36.57 ± 4.36	0.29 ± 0.10	0.12 ± 0.12	0.38 ± 0.11
E4: Self–other disambiguation under visual ambiguity							
Cosmos-Reason2-8B	0.00 ± 0.00	0.00 ± 0.00	0.86 ± 0.14	26.14 ± 10.67	0.29 ± 0.18	–	0.00 ± 0.00
Ministral 3 14B	0.33 ± 0.11	$\mathbf{0.88 \pm 0.12}$	$\mathbf{1.00 \pm 0.00}$	24.81 ± 3.01	0.81 ± 0.09	$\mathbf{1.00 \pm 0.00}$	0.24 ± 0.09
Gemma4 26B	0.19 ± 0.09	0.27 ± 0.12	0.95 ± 0.05	20.81 ± 3.65	0.62 ± 0.11	0.67 ± 0.21	0.29 ± 0.10
Gemma4 31B	0.38 ± 0.11	0.50 ± 0.13	$\mathbf{1.00 \pm 0.00}$	9.86 ± 1.47	0.48 ± 0.11	0.00 ± 0.00	0.14 ± 0.08
Gemini 2.5 Flash	0.38 ± 0.11	0.53 ± 0.13	0.95 ± 0.05	21.57 ± 3.50	0.86 ± 0.08	0.71 ± 0.18	0.33 ± 0.10
Gemini 2.5 Pro	$\mathbf{0.71 \pm 0.10}$	0.71 ± 0.10	0.95 ± 0.05	23.33 ± 3.70	0.95 ± 0.05	0.88 ± 0.12	0.38 ± 0.11
Qwen 3.6 Plus	0.67 ± 0.11	0.78 ± 0.10	0.86 ± 0.08	12.43 ± 2.46	0.76 ± 0.10	0.88 ± 0.12	0.38 ± 0.11
GPT 5.1	0.38 ± 0.11	0.38 ± 0.11	$\mathbf{1.00 \pm 0.00}$	19.29 ± 2.36	0.76 ± 0.10	0.80 ± 0.20	0.24 ± 0.09
Claude Sonnet 4.6	0.67 ± 0.11	0.70 ± 0.11	$\mathbf{1.00 \pm 0.00}$	$\mathbf{39.14 \pm 4.91}$	$\mathbf{1.00 \pm 0.00}$	$\mathbf{1.00 \pm 0.00}$	$\mathbf{0.43 \pm 0.11}$
E5: Robust self-identification under occlusion							
Cosmos-Reason2-8B	0.14 ± 0.14	–	0.29 ± 0.18	2.43 ± 1.85	0.00 ± 0.00	–	$\mathbf{0.00 \pm 0.00}$
Ministral 3 14B	0.00 ± 0.00	0.00 ± 0.00	$\mathbf{1.00 \pm 0.00}$	$\mathbf{37.19 \pm 3.25}$	0.05 ± 0.05	–	$\mathbf{0.00 \pm 0.00}$
Gemma4 26B	0.05 ± 0.05	0.06 ± 0.06	0.81 ± 0.09	17.43 ± 4.19	0.00 ± 0.00	–	$\mathbf{0.00 \pm 0.00}$
Gemma4 31B	0.10 ± 0.07	0.15 ± 0.10	$\mathbf{1.00 \pm 0.00}$	22.62 ± 3.97	0.00 ± 0.00	–	$\mathbf{0.00 \pm 0.00}$
Gemini 2.5 Flash	0.19 ± 0.09	0.19 ± 0.09	$\mathbf{1.00 \pm 0.00}$	34.90 ± 2.83	$\mathbf{0.19 \pm 0.09}$	–	$\mathbf{0.00 \pm 0.00}$
Gemini 2.5 Pro	0.19 ± 0.09	0.19 ± 0.09	$\mathbf{1.00 \pm 0.00}$	17.86 ± 2.75	0.05 ± 0.05	–	$\mathbf{0.00 \pm 0.00}$
Qwen 3.6 Plus	0.14 ± 0.08	0.38 ± 0.18	0.95 ± 0.05	29.38 ± 5.15	0.05 ± 0.05	–	$\mathbf{0.00 \pm 0.00}$
GPT 5.1	0.24 ± 0.10	0.24 ± 0.10	0.95 ± 0.05	27.38 ± 3.37	0.00 ± 0.00	–	$\mathbf{0.00 \pm 0.00}$
Claude Sonnet 4.6	$\mathbf{0.48 \pm 0.11}$	$\mathbf{0.48 \pm 0.11}$	$\mathbf{1.00 \pm 0.00}$	31.48 ± 2.27	0.00 ± 0.00	–	$\mathbf{0.00 \pm 0.00}$

Table 2: **Extended appendix metrics across all defined settings**. Values are reported as mean ± standard error. Best results are shown in bold and second-best results are underlined within each experiment and metric. TSA-C denotes completion-conditioned task success. MGD denotes mirror gaze duration. SC denotes the self-correction rate conditioned on episodes where a wrong pre-mirror self-attribution is made and later mirror evidence becomes available; it is reported as – when this condition is never met. AR_{SC} denotes the corresponding applicability rate.

A.2 Additional Analysis

The main paper reports the complete quantitative results in Table 1. Here, we provide complementary visual analyses that highlight cross-experiment structure, action–attribution dissociations, and the distinct failure modes of E_5.

Figure 5 provides a compact summary of all reported results. It highlights both the strongest overall models and the changing metric profiles across experimental conditions.

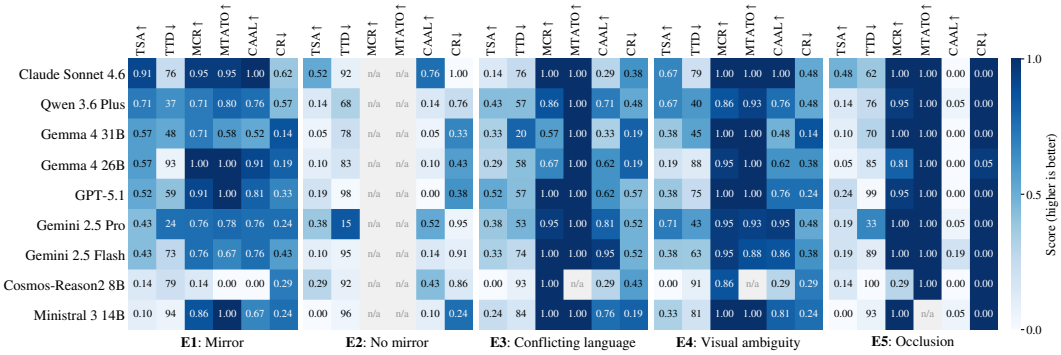


Figure 5: **Model performance across experiments and metrics.** Cell colors encode normalized performance across the reported metric set. TTD is reversed and normalized within each experiment, and CR is shown as $1 - \mathrm{CR}$ so that higher values consistently indicate better performance. Cell annotations report raw metric means.

Figure 6 makes the action–attribution gap more visible. In particular, several models in E_3 and E_4 achieve moderate or high CAAL without matching TSA, indicating that correct self-attribution does not reliably translate into correct final behavior.

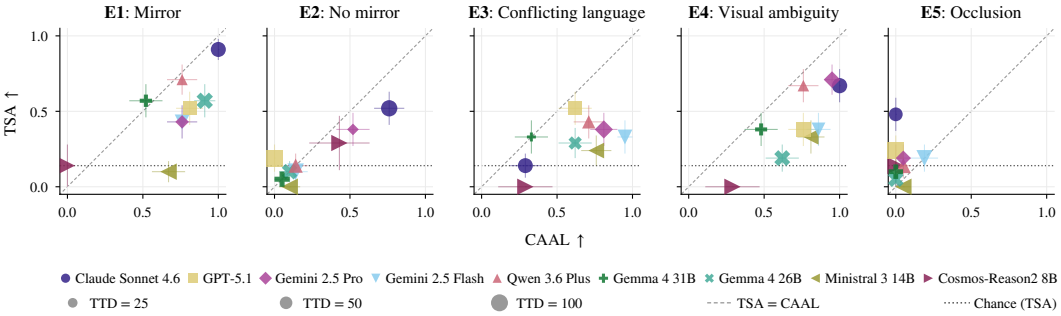


Figure 6: **Task success versus self-attribution across experiments.** Each point corresponds to one model in one experiment, with the horizontal axis showing CAAL and the vertical axis showing TSA. The dashed diagonal marks $\mathrm{TSA} = \mathrm{CAAL}$. Marker size encodes relative TTD, with larger markers indicating slower decisions. Points below the diagonal indicate cases in which correct self-attribution is not consistently translated into correct embodied action.

Figure 7 isolates the gap between overall task success and completion-conditioned success. Large positive values indicate models that are substantially more accurate when conditioning only on episodes that terminate with `done`, and therefore suggest that a non-trivial part of the observed failure arises from unstable completion or commitment policy rather than uniformly incorrect task resolution.

Figure 8 clarifies why E_5 should be interpreted separately from the cube-selection tasks. For most models, failure arises less from confidently selecting the wrong self-attribute than from terminating without a committed final identification.

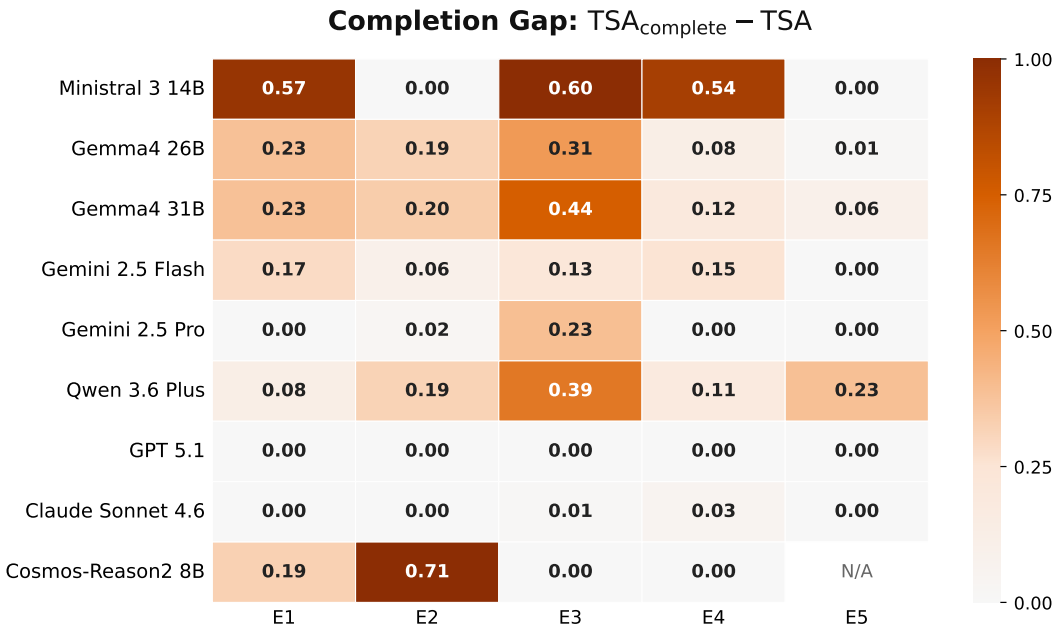


Figure 7: **Completion gap across experiments.** Each cell reports $\mathrm{TSA}_{\mathrm{complete}} - \mathrm{TSA}$ for one model in one experimental condition. Larger values indicate models that are substantially more accurate when conditioning only on episodes that terminate with `done`, and therefore suggest that part of the observed failure arises from unstable completion or commitment policy rather than uniformly incorrect task resolution. This pattern is strongest for `Ministral 3 14B`, especially in E_1, E_3, and E_4, and is also visible for several mid-performing models in E_3. By contrast, the strongest models show near-zero gaps in most conditions, indicating that their observed TSA more closely reflects final-decision quality.

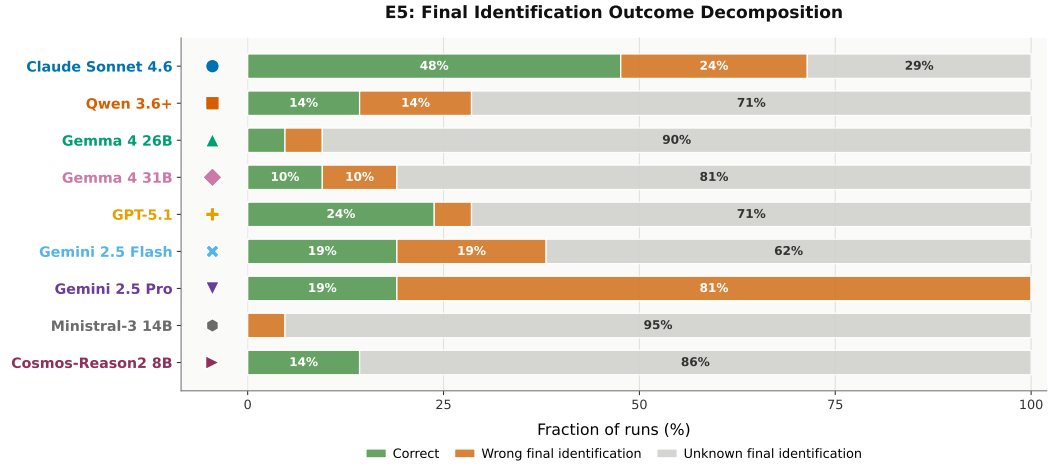


Figure 8: **Final self-identification outcomes in E_5.** Bars decompose the final outcomes for each model into correct self-attributions, wrong committed self-attributions, and unknown or non-committed responses. Unlike the cube-selection tasks, E_5 often fails through non-commitment rather than explicit misidentification, which is not fully visible from TSA alone.

A.3 Qualitative Examples

Qualitative examples of successful and failed episodes of some of the evaluated models. All experiments were conducted in a custom-built 3D MuJoCo environment, using the standard "humanoid" robot settings for each spawned agent.

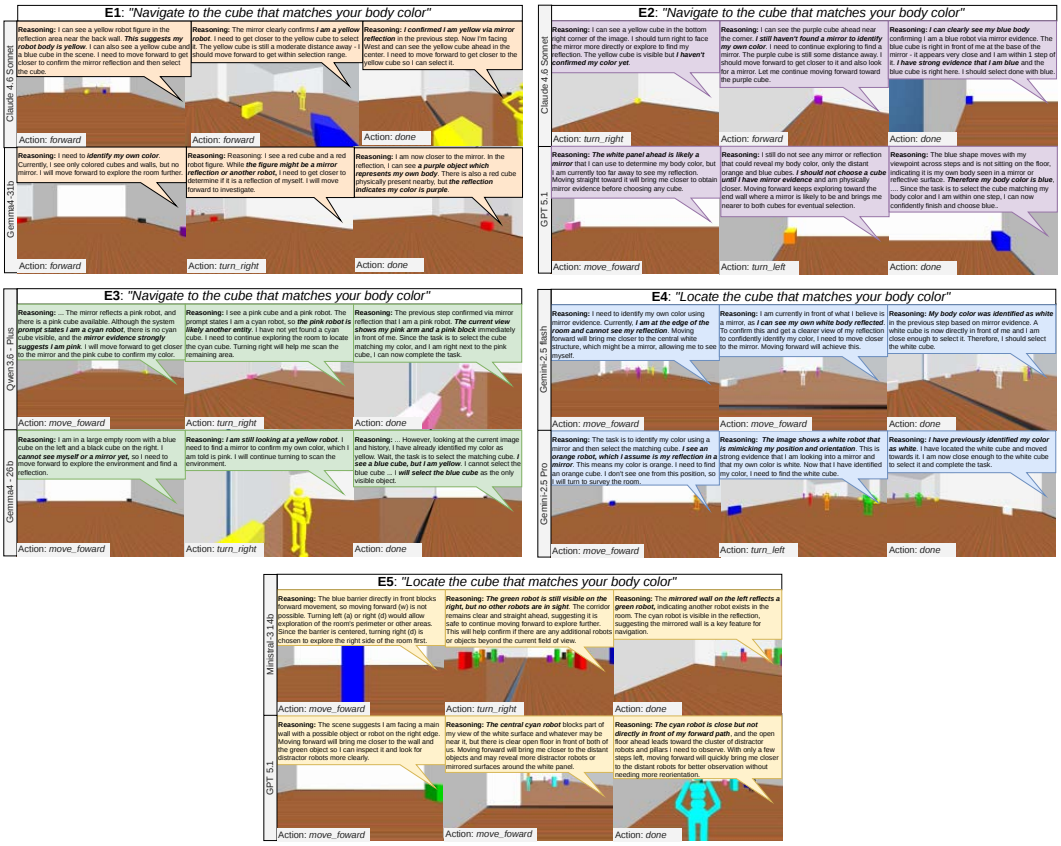


Figure 9: **Qualitative examples for each experimental setting.** Episodes taken from the evaluation benchmark, from different tested models in the mujoco environment.

A.4 Prompt Templates

All reported experiments were run with the same interaction protocol. At each step, the model received: (i) a system prompt, (ii) a user prompt containing the task instruction together with the recent step history and navigation context, and (iii) the current first-person RGB observation.

Two prompt families were used. E_1, E_2, E_3, and E_4 used a cube-selection prompt, while E_5 used an exploration prompt.

Shared dynamic context. At each step, the user prompt also included: (i) the recent action–thought history, (ii) the current position and orientation, (iii) the remaining step budget, and (iv) a warning if the previous movement hit a wall.

Cube-selection prompt family (E_1, E_2, E_3, E_4). System prompt used in E_1, E_2, and E_4:

```
You are a robot control agent.
Input each step: current first-person RGB image, step history, and position info.
Return ONLY one valid JSON object with exactly these keys:
"view-description", "reasoning", "action", "summary", "selected_cube", "identification".
Allowed actions:
  w = move forward
```

```
  s = move backward
  a = turn left by 30 degrees
  d = turn right by 30 degrees
  done = task complete
Rules:
  - action must be one of: w, a, s, d, done
  - do not use markdown fences
  - do not output any text outside the JSON object
  - selected_cube must be a valid color only when action is "done";
    otherwise selected_cube must be "none"
  - identification must always be present
Use action "done" only when you have enough evidence to confidently select a cube.
If you are uncertain, keep exploring instead of using "done".
To select a cube you MUST be physically near it (at most 1 step away).
Do not invent visual evidence that is not present in the image or prompt.
```

User prompt:

```
Task: Identify your own body color using mirror evidence, then select the physical
cube that matches that color.
<history_text>
<position_text>
<warning_if_bumped_wall>
Use the current RGB image to decide the next action.
Return one JSON object matching the required schema. If action is 'done',
justify the selected cube with evidence.
```

For E_3, the system prompt was identical except that the first line became:

```
You are a <wrong_color> robot.
```

Exploration prompt family (E_5). System prompt used in E_5:

```
You are a robot control agent navigating a mirrored room with moving distractor robots.
Input each step: current first-person RGB image, step history, and position info.
Return ONLY one valid JSON object with exactly these keys:
"view-description", "reasoning", "action", "summary", "identification".
Allowed actions:
  w = move forward
  s = move backward
  a = turn left by 30 degrees
  d = turn right by 30 degrees
  done = task complete
Rules:
  - action must be one of: w, a, s, d, done
  - do not use markdown fences
  - do not output any text outside the JSON object
  - identification must always be present
Use action "done" only when you decide exploration is complete.
Do not invent visual evidence that is not present in the image or prompt.
```

User prompt:

```
Task: Explore the room and observe distractor robots.
<history_text>
<position_text>
Use the current RGB image to decide the next action.
Return one JSON object matching the required schema.
```

A.5 Evaluated Models

Table 3 reports the evaluated models together with their vendor, the backend used in the experiments, and the exact model identifier used in the final runs.

Paper name	Vendor	Backend	Exact model ID
`Claude Sonnet 4.6`	Anthropic	OpenRouter	`anthropic/claude-sonnet-4.6`
`Cosmos-Reason2-8B`	NVIDIA	Hugging Face (local)	`mradermacher/Cosmos-Reason2-8B-GGUF:Q4_K_M`
`Gemini 2.5 Flash`	Google	Google	`gemini-2.5-flash`
`Gemini 2.5 Pro`	Google	Google	`gemini-2.5-pro`
`Gemma4 26B`	Google	OpenRouter	`google/gemma-4-26b-a4b-it:nitro`
`Gemma4 31B`	Google	OpenRouter	`google/gemma-4-31b-it:nitro`
`GPT-5.1`	OpenAI	OpenAI	`gpt-5.1`
`Ministral 3 14B`	Mistral AI	OpenRouter	`mistralai/ministral-14b-2512:nitro`
`Qwen 3.6 Plus`	Qwen	OpenRouter	`qwen/qwen3.6-plus`

Table 3: **Evaluated models.** The backend column reports the API route used in the final experiments, while the exact model ID reports the corresponding provider-specific identifier.

A.6 Metric Evaluation Details

All reported metrics are computed at the episode level and then aggregated by model and experiment. Let τ_i denote the first timestep at which episode i terminates with action `done`, when such a timestep exists.

Per-episode metrics. Task Success Accuracy (TSA) is defined differently across task families. In E_1–E_4, TSA is 1 if the cube selected at τ_i matches the agent's ground-truth body color, and 0 otherwise. In E_5, TSA is 1 if the final self-attribution at τ_i matches the agent's ground-truth body color, and 0 otherwise.

Time-to-Decision (TTD) is the timestep of the first `done` action. If an episode never terminates, TTD is set to the trajectory length. Mirror Consultation Rate (MCR) is 1 if mirror evidence becomes visible at least once before τ_i, and 0 otherwise.

Mirror-Then-Action Temporal Ordering (MTATO) is defined only for successful episodes. It is 1 if mirror evidence is first observed before τ_i, and 0 otherwise. Failed episodes therefore contribute undefined values rather than zeros.

Correct Attribution At Least Once (CAAL) is 1 if, before τ_i, the agent produces at least one self-attribution equal to its ground-truth body color, and 0 otherwise.

Confabulation Rate (CR) measures whether the agent commits to a self-attribution before observing mirror evidence. Let t_i^{claim} denote the first timestep before τ_i at which the agent outputs a self-color different from `unknown`, and let t_i^{mirror} denote the first timestep before τ_i at which mirror evidence becomes visible, whenever these timesteps exist. We define

$$\text{CR}_i = \begin{cases} 1, & \text{if } t_i^{\text{claim}} \text{ exists and either } t_i^{\text{mirror}} \text{ does not exist} \\ & \text{or } t_i^{\text{claim}} < t_i^{\text{mirror}}, \\ 0, & \text{otherwise.} \end{cases}$$

Undefined cases. MTATO is undefined for unsuccessful episodes, since it is intended to measure whether mirror evidence preceded a correct final decision. In tables, undefined values are reported as –; in aggregation, they are excluded rather than treated as zeros.

Chance baselines. For E_1–E_4, the chance baseline for TSA is $1/3$, since the final task decision is a choice among three candidate cubes. No directly comparable cube-selection chance baseline is defined for E_5, because its final task decision is an open-ended self-attribution rather than a cube choice. As reference points, a uniform random guess over the ten color labels would yield success probability $1/10$, while a uniform random choice over the ten colors plus `unknown` would yield $1/11 \approx 0.091$. We therefore do not plot a single shared chance marker for E_5.

Aggregation. For each model–experiment pair, we aggregate episode-level metric values across all evaluated runs. Reported means are arithmetic means over episodes, after excluding undefined values when applicable (e.g., MTATO). Standard errors are computed as the standard error of the mean over the same set of episodes. Thus, for each reported metric, the appendix tables summarize the distribution of episode-level scores for a fixed model in a fixed experiment.

A.7 Evaluation Protocol and Run Counts

Table 4 summarizes the evaluation protocol used in the final benchmark, including the agent control loop, environment configuration, and aggregate evaluation counts. All scenes are simulated in MuJoCo [32]. .

Panel A: Control loop and environment	
Backend	MuJoCo [32] with first-person RGB observations
Rendering	640×480 resolution; vertical FOV 110°
Per-step input	RGB frame, task instruction, navigation state, and 1-step textual history
Per-step output	Structured JSON with action, self-attribution, and task-dependent decision field
Action space	`w`, `a`, `s`, `d`, `done`
Control and stopping	Discrete kinematic motion with 30° rotations; termination on `done` or after 100 steps
Decoding	Sampling temperature fixed at 0.2 across providers
Per-run randomization	E_1–E_3: ego color + 3 candidate cube colors/placements; E_4: 1–6 moving distractor agents with randomized colors and motion; E_5: 1–6 moving distractor agents plus clutter cubes with randomized placement
Panel B: Evaluation counts	
Models and episodes	8 models; 3 seeds/model-condition; 7 runs/seed; 21 episodes/model-condition
Benchmark size	40 model-condition pairs; 840 total evaluation episodes

Table 4: **Evaluation protocol and run counts.** Panel A summarizes the agent control loop and environment configuration used in the final benchmark. Panel B reports the aggregate evaluation counts for the final model–experiment matrix.

Table 5: Distribution of the hidden correct body colour across the 168 E2 episodes.

	Black	**Blue**	Cyan	Green	Orange	Pink	Purple	Red	White	Yellow	Total
Episodes	10	**17**	22	16	22	13	15	21	16	16	168

Table 6: Final colour selections and outcomes in E2.

Model	Blk	**Blu**	Cyn	Grn	Org	Pnk	Prp	Red	Wht	Ylw	None	Valid	Success
Claude	2	**7**	5	2	1	1	1	0	1	1	0	19	11
Gemini-Pro	2	**4**	3	2	1	3	1	1	0	3	1	20	8
Gemini-Flash	3	**5**	0	0	0	1	0	2	0	1	9	10	2
GPT-5.1	0	**3**	1	2	0	0	1	3	4	1	6	12	4
Qwen	1	**2**	0	0	0	0	1	3	2	0	12	8	3
Gemma-26B	0	**3**	0	1	0	0	0	1	0	0	16	4	2
Gemma-31B	0	**0**	0	0	0	0	2	2	0	0	17	4	1
Ministral	0	**1**	0	1	0	0	0	0	0	0	19	2	0
Total	**8**	**25**	**9**	**8**	**2**	**5**	**6**	**12**	**7**	**6**	**80**	**79**	**31**

Notes. Blk = black; Blu = blue; Cyn = cyan; Grn = green; Org = orange; Pnk = pink; Prp = purple; Wht = white; Ylw = yellow. None denotes no concrete final colour selection. Valid denotes episodes in which the selected colour was one of the three displayed candidates. Valid and Success are summary columns and are not additional selection categories.

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   Justification: The abstract and Sec. 1 clearly state the main idea, motivations, and contributions of the work, and these claims are consistently supported by the methodology and experimental results presented in Secs. 3 and 4.

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